



CORPUS CHRISTI COLLEGE
SEQUERE DOMINUM

12 ATAR PHYSICS

2019 Program

matter particles

	1st gen.	2nd gen.	3rd gen.
QUARK	<i>u</i> <i>up</i>	<i>c</i> <i>charm</i>	<i>t</i> <i>top</i>
	<i>d</i> <i>down</i>	<i>s</i> <i>strange</i>	<i>b</i> <i>bottom</i>
LEPTON	<i>ν_e</i> <i>e neutrino</i>	<i>ν_μ</i> <i>μ neutrino</i>	<i>ν_τ</i> <i>τ neutrino</i>
	<i>e</i> <i>electron</i>	<i>μ</i> <i>muon</i>	<i>τ</i> <i>tau</i>

gauge particles

Strong Force <i>g</i> <i>gluon</i>
Electro-Magnetic Force <i>γ</i> <i>photon</i>
Weak Force <i>W⁺</i> <i>W bosons</i> <i>W⁻</i> <i>W bosons</i> <i>Z</i> <i>Z boson</i>

scalar particle(s)

<i>H</i> <i>Higgs</i> <i>?</i> <i>W bosons</i> <i>?</i> <i>Z boson</i> . . .

Elements of the Standard Model

Unit content

An understanding of the Year 11 content is assumed knowledge for students in Year 12. It is recommended that students studying Unit 3 and Unit 4 have completed Unit 1 and Unit 2.

This unit includes the knowledge, understandings and skills described below. This is the examinable content.

Science Inquiry Skills

- Identify, research and construct questions for investigation; propose hypotheses; and predict possible outcomes.
- Design investigations, including the procedure to be followed, the materials required, and the type and amount of primary and/or secondary data to be collected; conduct risk assessments; and consider research ethics.
- Conduct investigations, including the manipulation of force measurers and electromagnetic devices, safely, competently and methodically for the collection of valid and reliable data.
- Represent data in meaningful and useful ways, including using appropriate Système Internationale (SI) units, symbols, and significant figures; organise and analyse data to identify trends, patterns and relationships; identify sources of uncertainty and techniques to minimise these uncertainties; utilise uncertainty and percentage uncertainty to determine the uncertainty in the result of simple calculations, and evaluate the impact of measurement uncertainty on experimental results; and select, synthesise and use evidence to make and justify conclusions.
- Interpret a range of scientific and media texts, and evaluate processes, claims and conclusions by considering the accuracy and precision of available evidence; and use reasoning to construct scientific arguments.
- Select, construct and use appropriate representations, including text and graphic representations of empirical and theoretical relationships, vector diagrams, free body/force diagrams, field diagrams and circuit diagrams, to communicate conceptual understanding, solve problems and make predictions.
- Select, use and interpret appropriate mathematical representations, including linear and non-linear graphs and algebraic relationships representing physical systems, to solve problems and make predictions.
- Communicate to specific audiences and for specific purposes using appropriate language, nomenclature, genres and modes, including scientific reports.

12 ATAR Physics assessment table - Type of assessment		Weighting
Science Inquiry	There must be at least one experiment, one investigation and one evaluation and analysis completed in this pair of units. Appropriate strategies should be used to authenticate student achievement on an out-of-class assessment task.	
Experiment	Practical tasks designed to develop or assess a range of laboratory related skills and conceptual understanding of physics principles, and skills associated with representing data; organising and analysing data to identify trends and relationships; recognising error, uncertainty and limitations in data; and selecting, synthesising and using evidence to construct and justify conclusions. Tasks can take the form of practical skills tasks, laboratory reports and short in-class tests to validate the knowledge gained.	
Investigation	Activities in which ideas, predictions or hypotheses are tested and conclusions are drawn in response to a question or problem. Investigations can involve experimental testing, field work, locating and using information sources, conducting surveys, and using modelling and simulations. Assessment tasks can take the form of an experimental design brief, a formal investigation report requiring qualitative and/or quantitative analysis of the data and evaluation of physical information, or exercises requiring qualitative and/or quantitative analysis of second-hand data.	20%
Evaluation and analysis	Involves interpreting a range of scientific and media texts; evaluating processes, claims and conclusions by considering the accuracy and precision of available evidence; and using reasoning to construct scientific arguments. Assessment tasks can take the form of answers to specific questions based on individual research; exercises requiring analysis; and interpretation and evaluation of physics information in scientific and media texts.	
Test	Tests typically consist of questions requiring short answers, extended answers and problem solving. This assessment type is conducted in supervised classroom settings.	30%
Examination	Examinations require students to demonstrate use of terminology, understanding and application of concepts and knowledge of factual information. It is expected that questions would allow students to respond at their highest level of understanding. Typically conducted at the end of each semester and/or unit and reflecting the examination design brief for this syllabus. This assessment type is conducted in supervised classroom settings.	50%

Learning outcomes

By the end of this unit (U3), students:

- Understand that motion in gravitational, electric and magnetic fields can be explained using Newton's Laws of Motion.
- Understand how the electromagnetic wave model explains the production and propagation of electromagnetic waves across the electromagnetic spectrum.
- Understand transformations and transfer of energy in electromagnetic devices, as well as transformations and transfer of energy associated with motion in gravitational, electric and magnetic fields.
- Understand how models and theories have developed over time, and the ways in which physical science knowledge and associated technologies interact with social, economic, cultural and ethical considerations.
- Use science inquiry skills to design, conduct, analyse and evaluate investigations into uniform circular motion, projectile motion, satellite motion and gravitational and electromagnetic phenomena, and to communicate methods and findings.
- Use algebraic and graphical representations to calculate, analyse and predict measurable quantities related to motion, gravitational effects and electromagnetic phenomena.
- Evaluate, with reference to evidence, claims about motion, gravity and electromagnetic phenomena and associated technologies, and justify evaluations.
- Communicate physics understanding using qualitative and quantitative representations in appropriate modes and genres.

Weekly Lesson Outline - Term 4 (2018)

Wk	Topic	Text Reference	Homework
T4_4	Projectile motion - Independent nature of horizontal and vertical motion - Component nature of projectile motion - Time of flight (TOF) - Calculating speed in the horizontal and vertical plane	Pearson P46 – P62	STAWA Exploring physics set 3 Pearson review 2.2, 2.3 DOT point 1.5
The movement of free-falling bodies in Earth's gravitational field is predictable <i>This includes applying the relationship</i> $s = vt$, $v = u + at$, $s = ut + 1/2at^2$, $v^2 = u^2 + 2as$, $E_k = 1/2mv^2$			
T4_5	Parabola dynamics - The half parabola - The perfect parabola - The imperfect parabola - Quadratic nature of projectile motion	Pearson P46 – P62	STAWA Exploring physics set 3 Pearson review 2.2, 2.3
T4_6	Projectile motion calculations - Dynamics of parabolas - Range and air resistance - Speed of launch - Angle of launch	Pearson P46 – P62	DOT point 1.5
Projectile motion can be analysed quantitatively by treating the horizontal and vertical components of the motion independently <i>This includes applying the relationships</i> $s = vt$, $v = u + at$, $s = ut + 1/2at^2$, $v^2 = u^2 + 2as$, $E_k = 1/2mv^2$			
Projectile & Circular Motion Test (5%) - Complete			

Weekly Lesson Outline - Term 4 (2018)

Wk	Topic	Text Reference	Homework
T4_7	Circular Motion in the horizontal plane • Tension • Friction • Gravity Centripetal acceleration • Moving in a circle • Forces acting on a turning body Centripetal force in the horizontal plane • Equations of circular motion • Calculations with circular motion	Pearson P63 – P67	STAWA Exploring physics set 4 Pearson review 2.4 DOT point 1.6
	• When an object experiences a net force of const mag perpendicular to its velocity, it will undergo uniform circular motion, including circular motion on a horizontal plane and around a banked track; and vertical circular motion <i>This includes applying the relationships</i> $v = \frac{2\pi r}{T}, \quad a_c = \frac{v^2}{r}, \quad \text{resultant } F = m a_c = \frac{mv^2}{r}$		
Circular Motion Experiment (5%) - Complete			
T4_8	Centripetal force in the vertical plane Conical motion and Banked tracks	Pearson P63 – P86	STAWA Set 5, P71 DOT 1.9, 1.10, 1.11
	• When an object experiences a net force of const mag perpendicular to its velocity, it will undergo uniform circular motion, including circular motion on a horizontal plane and around a banked track; and vertical circular motion <i>This includes applying the relationships</i> $v = \frac{2\pi r}{T}, \quad a_c = \frac{v^2}{r}, \quad \text{resultant } F = m a_c = \frac{mv^2}{r}$		

Weekly Lesson Outline (2019)

Wk	Topic	Text Reference	Homework
T1_1	Revision of projectile motion & vectors	Pearson P40 – P62	STAWA Set 1, P19
T1_2	Revision of circular motion	Pearson P63 – P86	STAWA Set 4, P60 DOT 1.8, 1.9
	When an object experiences a net force of constant magnitude perpendicular to its velocity, it will undergo uniform circular motion, including circular motion on a horizontal plane and around a banked track; and vertical circular motion. <i>This includes applying the relationships</i> $v = \frac{2\pi r}{T}$, $a_c = \frac{v^2}{r}$, resultant $F = m a_c = \frac{mv^2}{r}$,		
T1_3	Newton's law of gravitation Gravitational fields	Pearson P4 – P37	STAWA Set 5, P71 DOT 1.9, 1.10, 1.11
	- All objects with mass attract one another with a gravitational force; the magnitude of this force can be calculated using Newton's Law of Universal Gravitation. <i>This includes applying the relationships</i> $F_g = G \frac{m_1 m_2}{r^2}$ - Objects with mass produce a gravitational field in the space that surrounds them; field theory attributes the gravitational force on an object to the presence of a gravitational field. <i>This includes applying the relationships</i> $F_{wt} = mg$ - When a mass moves or is moved from one point to another in a gravitational field and its potential energy changes, work is done on the mass by the field. <i>This includes applying the relationships</i> $E_p = m g \Delta h$, $W = F s$, $W = \Delta E$, $E_k = \frac{1}{2} m v^2$ - Gravitational field strength is defined as the net force per unit mass at a particular point in the field <i>This includes applying the relationships</i> $g = \frac{F_g}{m} = G \frac{M}{r^2}$		

Weekly Lesson Outline (2019)

Wk	Topic	Text Reference	Homework
T1_4	Satellites in orbit Kepler's laws Weightlessness	Pearson P87 – P96	STAWA Set 5, P71 DOT 1.9, 1.10, 1.11
Newton's Law of Universal Gravitation is used to explain Kepler's laws of planetary motion and to describe the motion of planets and other satellites, modeled as uniform circular motion <i>This includes deriving and applying the relationship</i> $\frac{T^2}{r^3} = \frac{4\pi^2}{GM}$			
T1_5	Centre of mass Torque	Pearson P98 – P108	STAWA Set 2, P37 DOT 1.12, 1.13
When an object experiences a net force at a distance from a pivot and at an angle to the lever arm, it will experience a torque or moment about that point <i>This includes deriving and applying the relationship</i> $\tau = r F \sin\theta$			
T1_6	Translational equilibrium Static equilibrium	Pearson P109 – P132	STAWA Set 2, P37
For a rigid body to be in equilibrium, the sum of the forces and the sum of the moments must be zero <i>This includes deriving and applying the relationship</i> $\sum F = 0$, $\tau = r F \sin\theta$, $\sum \tau = 0$			
T1_7	Static electricity and charge The Electric field, electric field lines Uniform electric fields	Pearson P134 – P138	STAWA Set 9, P124 DOT 4.1 & 4.2 (Yr11)
- The direction of conventional current is that in which the flow of positive charges takes place, while the electron flow is in the opposite direction. - A positively charged body placed in an electric field will experience a force in the direction of the field; the strength of the electric field is defined as the force per unit charge. <i>This includes applying the relationship</i> $E = \frac{F}{q}$			

Weekly Lesson Outline (2019)

Wk	Topic	Text Reference	Homework
T1_8	Coulombs law Work done in an electric field	Pearson P139 – P143 Pearson P144 – P149	STAWA Set 9, P124 DOT 4.3 & 4.4 (Yr11)
- Point charges and charged objects produce an electric field in the space that surrounds them; field theory attributes the electrostatic force on a point charge or charged body to the presence of an electric field. Electrostatically charged objects exert a force upon one another; the magnitude of this force can be calculated using Coulomb's Law. <i>This includes applying the relationship</i> $F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$ - When a charged body moves or is moved from one point to another in an electric field and its potential energy changes, work is done on the charge by the field. <i>This includes applying the relationship</i> $V = \frac{W}{q}$			
T1_9	The magnetic field	Pearson P152 – P180	STAWA Set 6, P88 DOT 2.1 to 2.6, 4.1 & 4.2
- Current-carrying wires are surrounded by magnetic fields; these fields are utilised in solenoids and electromagnets - The strength of the magnetic field produced by a current is a measure of the magnetic flux density <i>This includes applying the relationship</i> $B = \frac{\mu_0}{2\pi} \frac{I}{r}$			
T1_10	Forces on charged objects	Pearson P163 – P168	STAWA Set 10, P130 STAWA Set 11, P137
Magnets, magnetic materials, moving charges and current-carrying wires experience a force in a magnetic field when they cut flux lines; this force is utilised in DC electric motors and particle accelerators <i>This includes applying the relationships</i> $F = q v B$ where $v \perp B$, $F = I \ell B$ where $\ell \perp B$			
T2_1	DC Motors Commutators	Pearson P169 – P180	STAWA Set 6, P88 DOT 4.3, 4.4
The force due to a current in a magnetic field in a DC electric motor produces a torque on the coil in the motor <i>This includes applying the relationship</i> $\tau = r_{\perp} F$			

Weekly Lesson Outline (2019)

Wk	Topic	Text Reference	Homework
T2_2	Induced EMF Magnetic flux and Lenz law	Pearson P182 – P202	STAWA Set 7, P97 DOT 2.7, 2.8, 2.9
	- An induced EMF is produced by the relative motion of a straight conductor in a magnetic field when the conductor cuts flux lines - Magnetic flux is defined in terms of magnetic flux density and area <i>This includes applying the relationship</i> $EMF = l v B$ and $\Phi = B A_{\perp}$		
T2_3	Semester 1 Examinations		
T2_4	Semester 1 Examinations		
T2_5	Generators AC Quantities	Pearson P203 – P210	STAWA Set 7, P97 DOT 2.4, 2.9, 2.10
	A changing magnetic flux induces a potential difference; this process of electromagnetic induction is used in step-up and step-down transformers, DC and AC generators <i>This includes applying the relationship</i> Ind emf = $-N \frac{\Delta \Phi}{t}$, AC emf_{max} = $-2N l v B$ = $-2\pi N B A_{\perp} f$, $emf_{rms} = \frac{emf_{max}}{\sqrt{2}}$		
T2_6	Transformers Power Transmission	Pearson P210 – P224	STAWA Set 8, P113 DOT 2.11, 2.12
	Conservation of energy, expressed as Lenz's Law of electromagnetic induction, is used to determine the direction of induced current <i>This includes applying the relationship</i> $V_p N_s = V_s N_p$ and $P = IV = I^2 R = V^2/R$		

Learning outcomes

By the end of this unit (U4), students:

- Understand the consequences for space and time of the equivalence principle for inertial frames of reference
- Understand how the quantum theory of light and matter explains black body radiation, the photoelectric effect, and atomic emission and absorption spectra
- Use the Standard Model to describe the nature of and interaction between the fundamental particles that form the building blocks of matter
- Understand how models and theories have developed over time, and the ways in which physical science knowledge and associated technologies interact with social, economic, cultural and ethical consideration
- Use science inquiry skills to design, conduct, analyse and evaluate investigations into frames of reference, diffraction, black body and atomic emission spectra, the photoelectric effect, and photonic devices, and to communicate methods and findings
- Use algebraic and graphical models to solve problems and make predictions related to the theory and applications of special relativity and quantum theory
- Evaluate the experimental evidence that supports the theory of relativity, wave-particle duality, the Bohr model of the atom, the Standard Model, and the Big Bang theory
- Communicate physics understanding using qualitative and quantitative representations in appropriate modes and genres.

Weekly Lesson Outline (2019)

Wk	Topic	Text Reference	Homework
T2_7	<u>Properties of waves</u> <u>Electromagnetic waves</u>	Pearson P232 – P259	STAWA Set 12.1, P160
Electromagnetic waves are transverse waves made up of mutually perpendicular, oscillating E and B fields oscillating charges produce electromagnetic waves of the same frequency as the oscillation; electromagnetic waves cause charges to oscillate at the frequency of the wave			
T2_8	<u>Atomic spectra and the Bohr model</u> <u>Spectra & energy level diagrams</u> <u>Fluorescence, phosphorescence</u>	Pearson P271 – P291	STAWA Set 13, P180
- The Bohr model of the hydrogen atom integrates light quanta and atomic energy states to explain the specific wavelengths in the hydrogen spectrum and in the spectra of other simple atoms; the Bohr model enables line spectra to be correlated with atomic energy-level diagrams - Atoms of an element emit and absorb specific wavelengths of light that are unique to that element; this is the basis of spectral analysis. <i>This includes applying the relationships</i> $\Delta E = hf$, $E_2 - E_1 = hf$			
T2_9	<u>Waves and particles</u>	Pearson P292 – P301	STAWA Set 12.1, P160
- Light exhibits many wave properties; however, it cannot only be modelled as a mechanical wave because it can travel through a vacuum - A wave model explains a wide range of light-related phenomena, including reflection, refraction, dispersion, diffraction and interference; a transverse wave model is required to explain polarisation - On the atomic level, energy and matter exhibit the characteristics of both waves and particles. Young's double slit experiment is explained with a wave model but produces the same interference and diffraction patterns when one photon at a time or one electron at a time are passed through the slits			

Weekly Lesson Outline (2019)

Wk	Topic	Text Reference	Homework
T2_10	Quanta of light The photoelectric effect	Pearson P260 – P270	STAWA Set 12.2, P164
- Atomic phenomena and the interaction of light with matter indicate that states of matter and energy are quantised into discrete values - A wide range of phenomena, including black body radiation and the photoelectric effect, are explained using the concept of light quanta - At the atomic level, electromagnetic radiation is emitted or absorbed in discrete packets called photons. The energy of a photon is proportional to its frequency. The constant of proportionality, Planck's constant, can be determined experimentally using the photoelectric effect and the threshold voltage of coloured LEDs <i>This includes applying the relationships</i> $c = f \lambda$, $E = hf = \frac{hc}{\lambda}$, $E_k = hf - W$, de Broglie $\lambda = \frac{h}{p}$			
T3_1	Special Relativity	Pearson P304 – P316	STAWA Set 14.4, P200
- Einstein's special theory of relativity predicts significantly different results to those of Newtonian physics for velocities approaching the speed of light - The special theory of relativity is based on two postulates: that the speed of light in a vacuum is an absolute constant, and that all inertial reference frames are equivalent			
T3_2	Time dilation Length contraction Muon decay	Pearson P317 – P330	STAWA Set 14.2, P202
- Observations of objects travelling at very high speeds cannot be explained by Newtonian physics. These include the dilated half-life of high-speed muons created in the upper atmosphere, and the momentum of high-speed particles in particle accelerators - Motion can only be measured relative to an observer; length and time are relative quantities that depend on the observer's frame of reference. <i>This includes applying the relationships</i>			$\ell = \ell_0 \sqrt{\left(1 - \frac{v^2}{c^2}\right)}$ $t = \frac{t_0}{\sqrt{\left(1 - \frac{v^2}{c^2}\right)}}$ $u = \frac{v + u'}{1 + \frac{vu'}{c^2}}$ $u' = \frac{u - v}{1 - \frac{uv}{c^2}}$

Weekly Lesson Outline (2019)

Wk	Topic	Text Reference	Homework
T3_3	Mass and energy equivalence	Pearson P331 – P389	STAWA Set 14.2, P202
	- Relativistic momentum increases at high relative speed and prevents an object from reaching the speed of light - The concept of mass-energy equivalence emerged from the special theory of relativity and explains the source of the energy produced in nuclear reactions. <i>This includes applying the relationships</i>	$p_v = \frac{m v}{\sqrt{1 - \frac{v^2}{c^2}}}$	$E = \frac{m c^2}{\sqrt{1 - \frac{v^2}{c^2}}}$
T3_4	The Standard Model of particle physics Particles of the Standard Model Fundamental particles Composite particles	Pearson P342 – P349	STAWA Set 15.1, P220
	High-energy particle accelerators are used to test theories of particle physics, including the Standard Model <i>This includes applying the relationship</i> $\frac{m v^2}{r} = q v B$		
T3_5	Particle interaction	Pearson P350 – P357	STAWA Set 15.2, P221
	Lepton number and baryon number are examples of quantities that are conserved in all reactions between particles; these conservation laws can be used to support or invalidate proposed reactions. Baryons are composite particles made up of quarks		
T3_6	Particle detectors Particle accelerators	Pearson P358 – P369	STAWA Set 15.2, P221
	The Standard Model is based on the premise that all matter in the universe is made up from elementary matter particles called quarks and leptons; quarks experience the strong nuclear force, leptons do not		

Weekly Lesson Outline (2019)

Wk	Topic	Text Reference	Homework
T3_7	The Expansion of the Universe	Pearson P370 – P314	STAWA Set 15.2, P221
- The Standard Model explains three of the four fundamental forces (strong, weak and electromagnetic forces) in terms of an exchange of force-carrying particles called gauge bosons; each force is mediated by a different type of gauge boson - The expansion of the universe can be explained by Hubble's law and cosmological concepts, such as red shift and the Big Bang theory - The Standard Model is used to describe the evolution of forces and the creation of matter in the Big Bang theory			
T3_8	Semester Two Exam Revision		
T3_9	Semester Two Exam Revision		
T3_10	Semester Two Exam Revision		

Year 12 Physics – 2019

Assessment profile

Unit 3				Unit 4			
Assessment	Weight	Out of	Date	Assessment	Weight	Out of	Date
Science Inquiry				Science Inquiry			
Circular motion experiment	5%	54	Wk7_T4 (2018)	Experiment	5%	45	Wk7_T3
Evaluation and analysis	5%	45	Wk6_T1	Investigation	5%	45	Wk1_T3
Tests				Tests			
Projectile & circular motion	5%	47	Wk6_T4 (2018)	Atomic physics	5%	45	Wk9_T2
Gravity, satellites & moments	5%	45	Wk5_T1	Relativity	5%	45	Wk3_T3
Electromagnetism & EM induction	5%	45	Wk5_T2	Particles & cosmology	5%	45	Wk6_T3
Examinations				Examinations			
Semester One	22%	180	Wk3_T2	Semester Two	28%	180	Term 3